Dokerized Construction Services

Course Name: DevOps

Institution Name: Medi-caps University – Datagami Skill Based Course

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Project Number: DO-18

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Academic Year: 2025-26

Problem Statement & Objectives

1. Problem Statement

Modern construction service applications face significant challenges in deployment and scalability. Traditional deployment methods often result in:

- Inconsistent environments across development and production
- Dependency conflicts and configuration issues
- Time-consuming manual setup processes
- Difficulty in scaling applications for large projects

These issues reduce productivity and increase the chances of system failures. Therefore, a containerized solution is required to ensure consistent, scalable, and efficient deployment of construction services

2. Project Objectives

The main objectives of the Dockerized Construction Service are:

- To containerize the construction service using Docker
- To ensure consistency across development, testing, and production environments
- To optimize application performance using multi-stage Docker builds
- To automate build and deployment processes using CI/CD pipelines
- To enable scalability and high availability using container orchestration

3. Scope of the Project

The project focuses on:

- Containerizing the construction service application
- Managing multi-container environments using Docker Compose
- Automating deployment through CI/CD pipelines
- Supporting scalable deployments using Kubernetes
- Ensuring portability across different systems and platforms.

Limitations:

- Requires understanding of Docker and DevOps concepts
- Initial setup complexity for beginners
- Kubernetes deployment may require cloud infrastructure
- Debugging containerized applications can be challenging.

4. Proposed Solution

The proposed system is a Docker-based containerized construction service that ensures efficient deployment and scalability.

The solution includes:

- Creating optimized multi-stage Dockerfiles to reduce image size and improve security
- Using Docker Compose to manage application and database services
- Implementing CI/CD pipelines using GitHub Actions for automation
- Deploying containers using Kubernetes for scalability and fault tolerance

This approach ensures a reliable, consistent, and production-ready system.

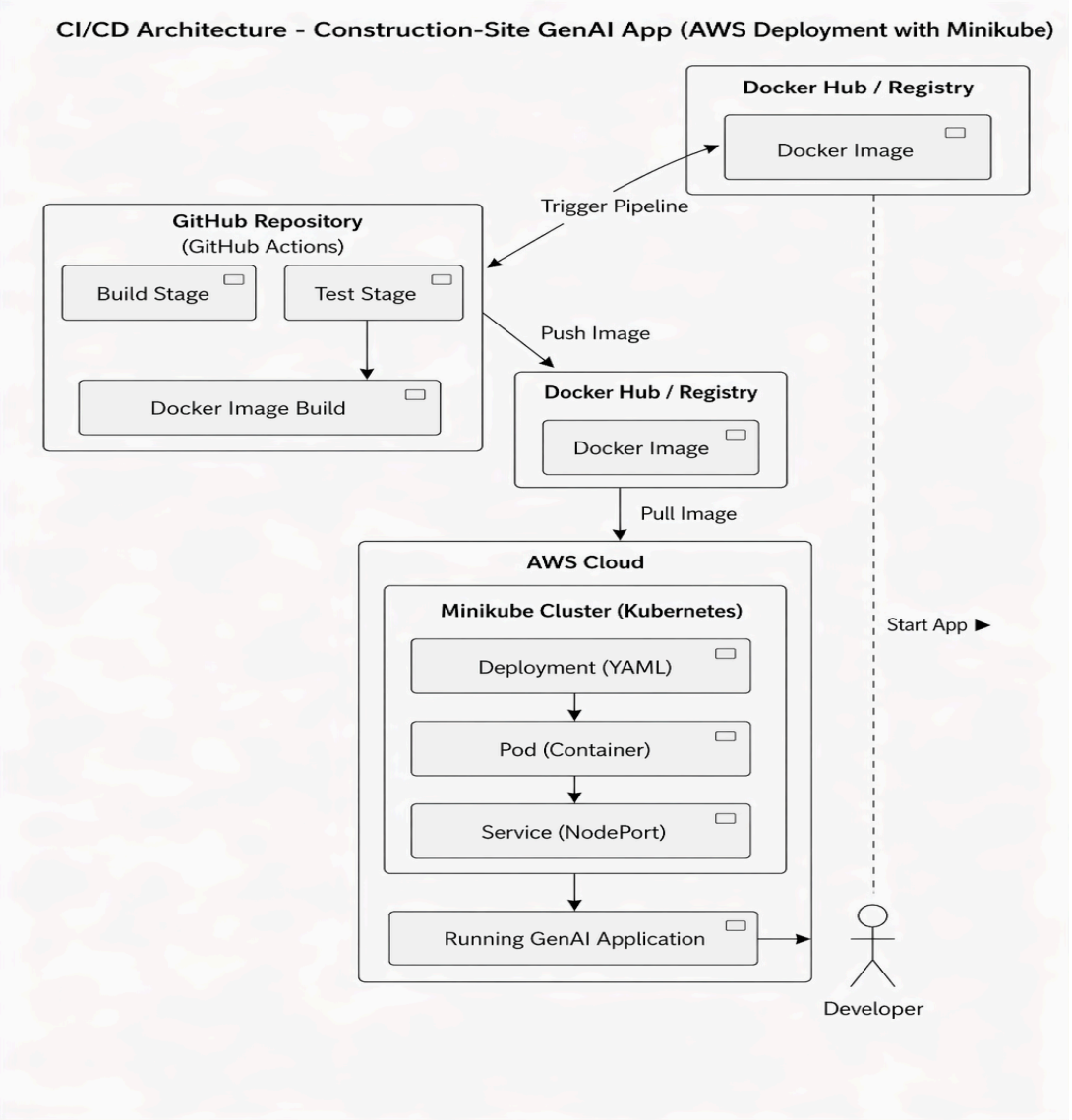
A. Key features

- Multi-stage Docker builds for optimized performance
- Container orchestration using Docker Compose
- Automated CI/CD pipeline integration
- Scalable deployment using Kubernetes
- Improved security and reduced image size
- Faster and consistent application deployment

B. Overall Architecture / Workflow

The system follows a structured DevOps workflow:

1. Developer writes code and pushes it to GitHub
2. CI/CD pipeline is triggered automatically
3. Docker image is built using a multi-stage Dockerfile
4. Image is stored in a container registry
5. Docker Compose or Kubernetes deploys the application
6. Users access the application through deployed services



C. Tools & Technologies Used

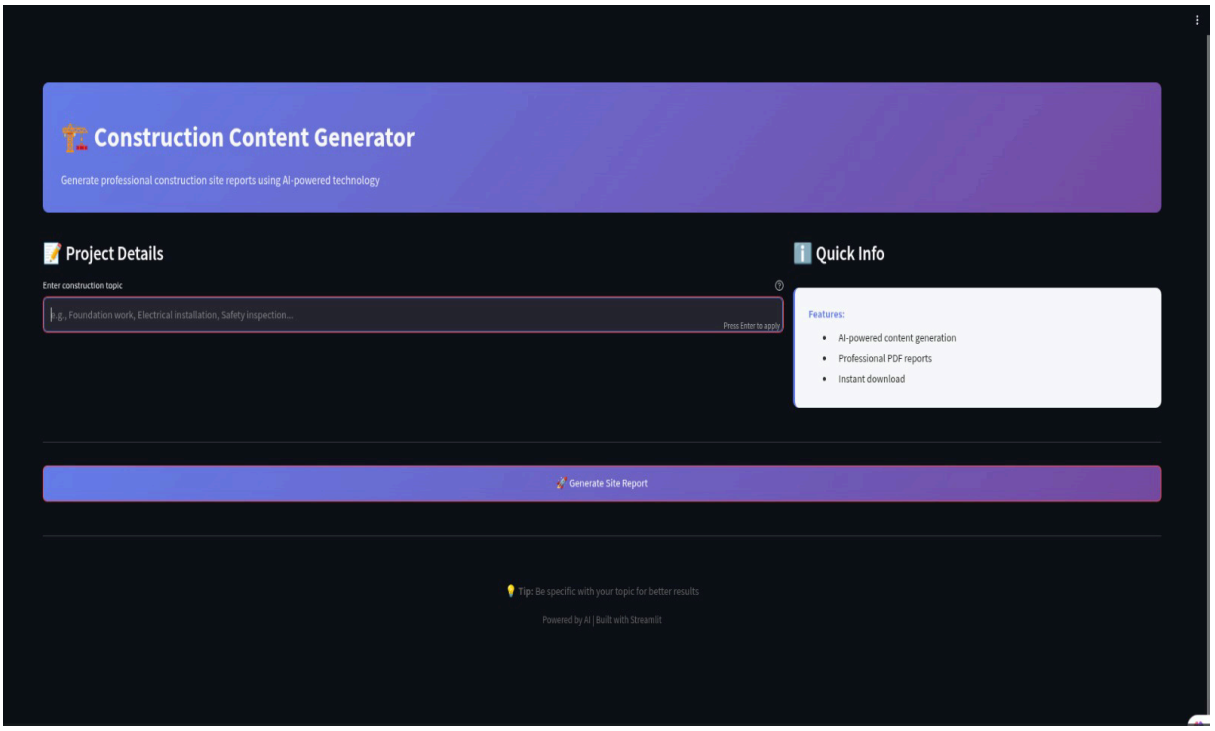
- Programming Language: Python
- Containerization: Docker
- CI/CD Tools: GitHub Actions
- Orchestration: Kubernetes
- Version Control: Git & GitHub

Results & Output

1. Screenshots / outputs

The project successfully demonstrates:

- Faster and automated deployment process
- Consistent environment across all stages
- Reduced manual configuration errors
- Improved scalability and system reliability


Construction Content Generator
Generate professional construction site reports using AI-powered technology
Project Details
Enter construction topic
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Press Enter to apply
Quick Info
Features:
AI-powered content generation
Professional PDF reports
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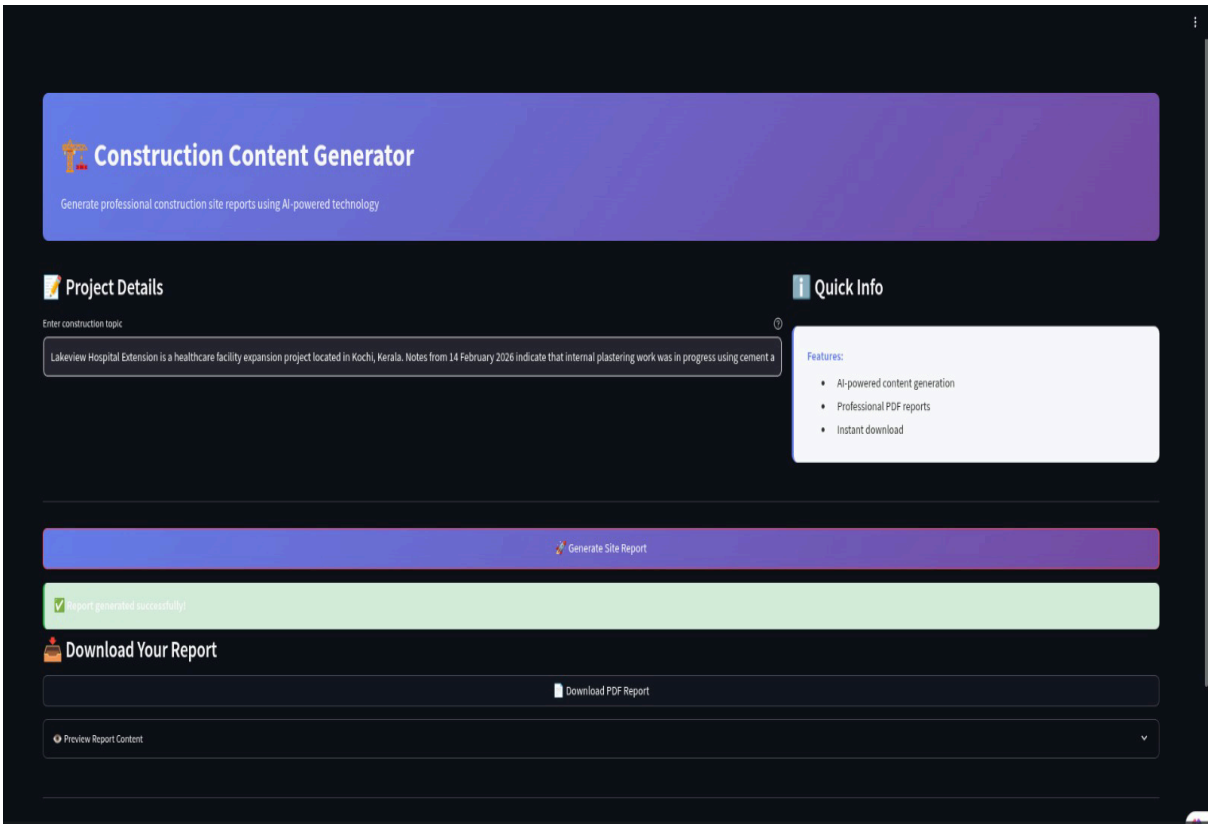
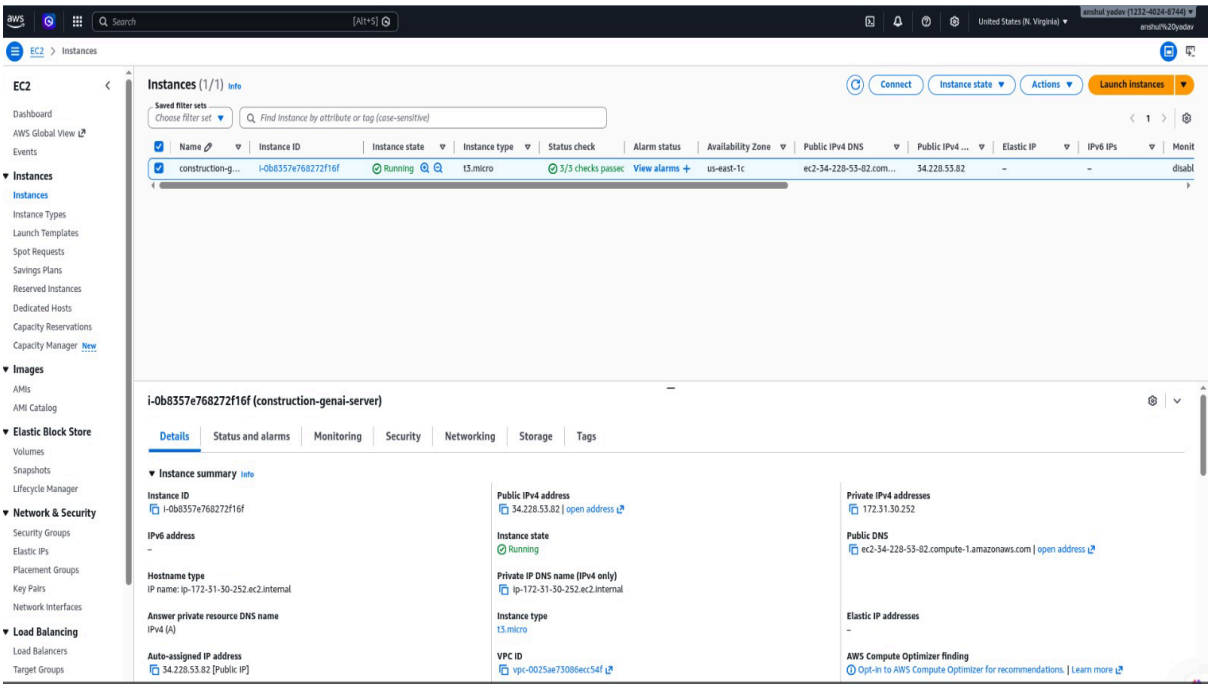
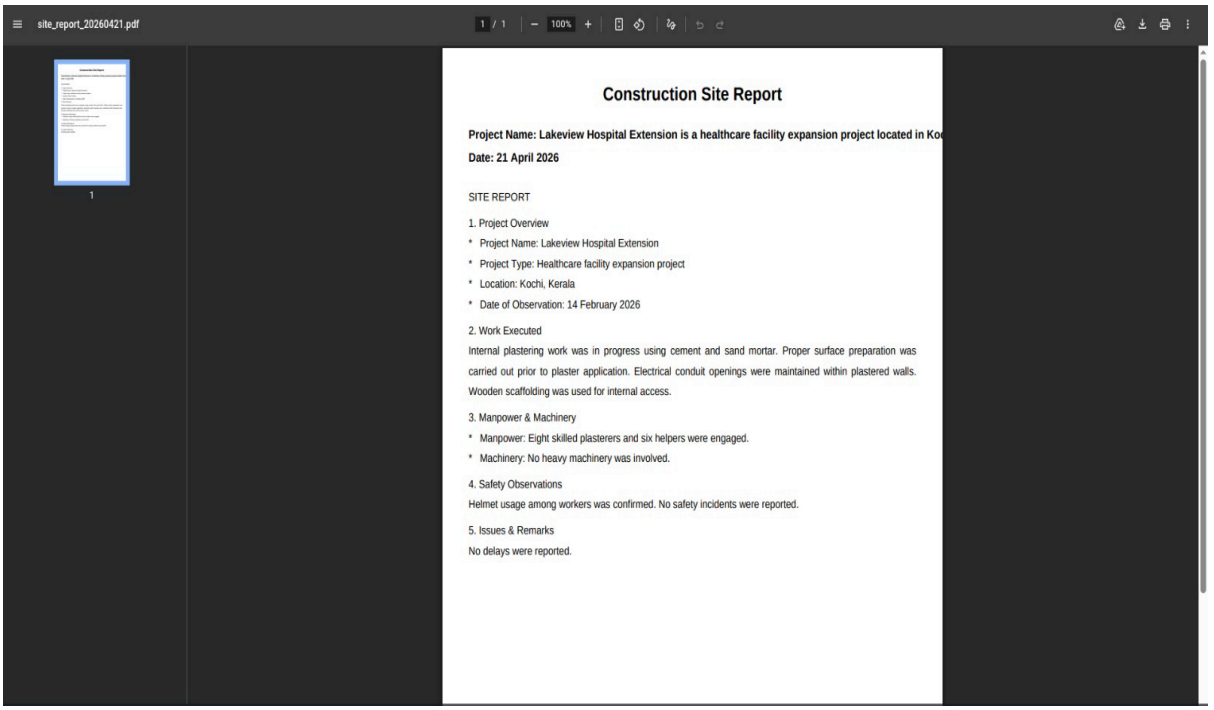
Generate Site Report
Tip: Be specific with your topic for better results
Powered by AI | Built with Streamlit


Construction Content Generator
Generate professional construction site reports using AI-powered technology
Project Details
Enter construction topic
Lakeview Hospital Extension is a healthcare facility expansion project located in Kochi, Kerala. Notes from 14 February 2026 indicate that internal plastering work was in progress using cement a
Quick Info
Features:
AI-powered content generation
Professional PDF reports
Instant download
Generate Site Report
Report generated successfully!
Download Your Report
Download PDF Report
Preview Report Content


EC2 > Instances
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
construction-g...
i-0b8357e768272f16f
Running
t3.micro
3/3 checks passed
View alarms +
us-east-1c
ec2-34-228-53-82.com...
34.228.53.82
i-0b8357e768272f16f (construction-genai-server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-0b8357e768272f16f
Public IPv4 address
34.228.53.82 | open address
Private IPv4 addresses
172.31.30.252
IPv6 address
Instance state
Running
Public DNS
ec2-34-228-53-82.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-30-252.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-30-252.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
34.228.53.82 [Public IP]
VPC ID
vpc-0025ae73086ecc54f
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more


site_report_20260421.pdf
Construction Site Report
Project Name: Lakeview Hospital Extension is a healthcare facility expansion project located in Ko
Date: 21 April 2026
SITE REPORT
1. Project Overview
* Project Name: Lakeview Hospital Extension
* Project Type: Healthcare facility expansion project
* Location: Kochi, Kerala
* Date of Observation: 14 February 2026
2. Work Executed
Internal plastering work was in progress using cement and sand mortar. Proper surface preparation was carried out prior to plaster application. Electrical conduit openings were maintained within plastered walls. Wooden scaffolding was used for internal access.
3. Manpower & Machinery
* Manpower: Eight skilled plasterers and six helpers were engaged.
* Machinery: No heavy machinery was involved.
4. Safety Observations
Helmet usage among workers was confirmed. No safety incidents were reported.
5. Issues & Remarks
No delays were reported.

2. **Key outcomes**

- Deployment time reduced significantly
- Increased development efficiency
- Improved system stability
- Better collaboration among team members

Conclusion

The Dockerized Construction Service project demonstrates how modern DevOps practices can transform application deployment.

The system:

- Eliminates environment inconsistencies
- Automates deployment workflows
- Enhances scalability and reliability

This project highlights the importance of containerization and automation in real-world software systems and provides practical experience in Docker, CI/CD, and orchestration tools.

Future Scope & Enhancements

- Deployment on cloud platforms (AWS, Azure, GCP)
- Integration with monitoring tools like Prometheus and Grafana
- Advanced security scanning for Docker images
- Implementation of auto-scaling policies in Kubernetes
- Integration with real-time construction management systems